

ROCHESTER PIER, NY, USA

WMO: 998009

Lat: 43.2625N

Lon: 77.5981W

Elev: 83

StdP: 100.33

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.3	(c) -11.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.4	(k) -3.8	(l) 12.9	(m) -5.4	(n) 6.5	(o) 210	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.8	(c) 30.0	(d) N/A	(e) 28.1	(f) N/A	(g) 26.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.6	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.5	(b) 11.2	(c) 10.0	(d) DB	(e) -16.0	(f) 32.7	(g) 2.4	(h) 1.0	(i) -17.7	(j) 33.5	(k) -19.1	(l) 34.1	(m) -20.5	(n) 34.6	(o) -22.3	(p) 35.3	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-2.9	-2.3	1.6	7.9	14.1	18.5	22.3	21.4	18.1	11.9	5.9	1.1
	DBStd	9.80	5.87	5.43	4.98	4.40	4.32	3.71	2.88	2.39	3.67	3.79	4.74	4.75
	HDD10.0	1556	401	346	266	93	9	0	0	0	0	23	139	278
	HDD18.3	3419	657	578	520	315	146	43	2	3	47	201	373	534
	CDD10.0	1509	2	2	4	30	136	254	382	353	244	83	16	3
	CDD18.3	332	0	0	0	2	15	48	126	97	41	3	0	0
	CDH23.3	1573	0	0	0	14	96	244	652	356	198	13	0	0
	CDH26.7	338	0	0	0	3	19	62	144	57	49	2	0	0
(14) Wind	WSAvg	4.8	5.7	5.7	5.6	5.3	4.1	3.9	3.8	4.0	4.3	5.1	5.1	5.3
(15) Precipitation	PrecAvg	844	58	51	61	78	79	73	83	80	75	82	63	61
	PrecMax	1069	145	101	121	147	148	147	186	130	132	152	100	114
	PrecMin	682	26	16	20	26	21	8	27	25	31	27	15	27
	PrecStd	91	28	20	25	31	34	40	45	26	29	35	23	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	16.2	18.0	26.5	29.1	31.3	32.5	30.6	31.1	24.8	20.1	16.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	12.2	11.6	15.1	21.5	25.9	28.5	30.0	28.2	27.8	21.3	17.6	12.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.2	7.7	11.8	18.5	23.3	26.3	28.1	26.5	25.3	19.0	15.2	10.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	4.9	4.2	8.3	15.3	20.7	23.9	26.7	25.2	23.6	17.4	12.8	7.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.2	6.2	7.4	7.9	6.3	5.8	5.8	6.4	6.2	6.0	5.3
		MCDBR	10.3	11.5	11.6	13.9	11.5	9.3	7.7	7.6	8.5	8.9	9.4	8.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.335	0.373	0.397	0.439	0.435	0.423	0.378	0.360	0.316	0.296
	taud		2.396	2.370	2.394	2.344	2.329	2.248	2.270	2.288	2.397	2.428	2.511	2.491
	Ebn at Noon		821	878	906	893	877	837	835	833	848	818	813	810
	Edh at Noon		78	94	104	119	124	135	131	125	104	89	70	65
(44) All-Sky Solar Radiation	RadAvg		1.28	2.06	3.36	4.50	5.52	5.86	5.94	5.28	4.19	2.52	1.54	1.01
	RadStd		0.12	0.20	0.32	0.48	0.56	0.47	0.45	0.29	0.35	0.23	0.20	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page